SGM2045

300mA, Low V_{IN}, Ultra-Low Noise, High PSRR, Low Dropout Linear Regulator

SG Micro Corp — Datasheet & Engineering Specification

GENERAL DESCRIPTION

The SGM2045 is an ultra-low noise, high power supply rejection ratio (PSRR), low-dropout (LDO) CMOS linear regulator designed for RF and noise-sensitive wireless applications. Operating from an input voltage range of 1.1V to 5.5V, it delivers up to 300mA output current. The device features an exceptional 92dB PSRR at 1kHz and an ultra-low output voltage noise of 9.5μV_{RMS} (10Hz to 100kHz), making it ideal for powering satellite navigation receivers (GNSS), external low-noise amplifiers (LNAs), sensitive RF front-ends, and analog audio circuits.

The SGM2045 incorporates an internal P-channel MOSFET pass transistor, providing an ultra-low dropout voltage of 100mV (TYP) at 300mA in XTDFN-1×1-4L (and 80mV in WLCSP). At typical GNSS subsystem loads of 48.5mA (ATGM336H + ATR2652), the measured dropout voltage is merely 15mV to 20mV, drastically minimizing linear thermal dissipation. The device features an active logic-controlled shutdown (EN) with automatic output discharge that pulls the quiescent current down to 30nA (0.03μA TYP), short-circuit current limiting, and thermal overload protection.

FEATURES

- **Operating Input Voltage Range:** 1.1V to 5.5V
- **Fixed Output Voltage Options:** 0.6V to 4.2V / 4.5V (3.3V Fixed Version: SGM2045-3.3)
- **Output Current Capability:** 300mA continuous
- **High Output Voltage Accuracy:** ±1.0% at +25°C
- **Ultra-High Power Supply Rejection Ratio (PSRR):** 92dB at 1kHz, 75dB at 10kHz, 55dB at 100kHz
- **Ultra-Low Output Voltage Noise:** 9.5μV_{RMS} (10Hz to 100kHz)
- **Ultra-Low Dropout Voltage:** 100mV (TYP) at 300mA (XTDFN-1×1-4L); ~15mV to 20mV at 48.5mA load
- **Low Quiescent Current:** 15μA (TYP) at no load; **Shutdown Leakage:** 0.03μA (TYP) / 30nA
- **Active Output Discharge:** Fast discharge of output rail when disabled
- **Robust Protection:** Thermal shutdown protection (160°C) and foldback current limit
- **Miniature Packaging:** Green XTDFN-1×1-4L (1.0mm × 1.0mm) and WLCSP-0.64×0.64-4B-A
- **Operating Junction Temperature:** -40°C to +125°C

PIN CONFIGURATION & DESCRIPTION (XTDFN-1×1-4L)

Pin	Name	I/O	Description
1	OUT	O	Regulated Output Voltage. Bypass with a 1μF low-ESR ceramic capacitor to GND.
2	GND	G	Ground. Connect to PCB ground plane.
3	EN	I	Enable Input Pin. Logic HIGH enables the device; logic LOW disables the device and initiates active output discharge. Do not leave floating.
4	IN	P	Input Supply Voltage (1.1V to 5.5V). Bypass with a 1μF ceramic capacitor to GND.
EP	GND / EP	G	Exposed thermal pad. Internally connected to GND. Solder to PCB ground plane.

ABSOLUTE MAXIMUM RATINGS & RECOMMENDED OPERATING CONDITIONS

Parameter	Symbol	Value	Unit
Input Voltage Range	V_{IN}	-0.3 to 6.0	V

Enable Voltage Range	V_{EN}	-0.3 to 6.0	V
Output Voltage Range	V_{OUT}	-0.3 to (V_{IN} + 0.3) < 6.0	V
Output Short-Circuit Duration	t_{SC}	Continuous	-
Junction Temperature	T_J	+150	°C
Storage Temperature Range	T_{STG}	-65 to +150	°C
Operating Junction Temperature	T_{OPR}	-40 to +125	°C
Recommended Input Voltage	V_{IN}	1.1 to 5.5	V
Recommended Output Capacitor	C_{OUT}	1.0 to 10 (Ceramic, X5R/X7R)	µF

ELECTRICAL CHARACTERISTICS

V_{IN} = $V_{OUT(NOM)}$ + 0.5V or 2.2V (whichever is greater), V_{EN} = V_{IN}, C_{IN} = C_{OUT} = 1μF, T_A = +25°C unless otherwise noted.

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Input Voltage	V_{IN}	-	1.1	-	5.5	V
Output Voltage Accuracy	V_{OUT}	I_{OUT} = 1mA, T_A = +25°C	-1.0	-	+1.0	%
Maximum Output Current	I_{MAX}	Continuous	300	-	-	mA
Current Limit	I_{LIM}	V_{OUT} = 90% × $V_{OUT(NOM)}$	350	450	650	mA
Quiescent Current	I_Q	I_{OUT} = 0mA, V_{EN} = V_{IN}	-	15	25	μA
Shutdown Current	I_{SHDN}	V_{EN} = 0V, V_{IN} = 4.2V	-	0.03	0.1	μA
Dropout Voltage (XTDFN)	V_{DROP}	I_{OUT} = 300mA, V_{OUT} = 3.3V	-	100	150	mV
Dropout Voltage (Light Load)	V_{DROP}	I_{OUT} = 48.5mA (GNSS Subsystem)	-	16	25	mV
Line Regulation	ΔV_{LINE}	V_{IN} = (V_{OUT} + 0.5V) to 5.5V, I_{OUT} = 1mA	-	0.01	0.05	%/V
Load Regulation	ΔV_{LOAD}	I_{OUT} = 1mA to 300mA	-	5	15	mV
Power Supply Rejection Ratio	PSRR	f = 1kHz, I_{OUT} = 50mA f = 10kHz, I_{OUT} = 50mA f = 100kHz, I_{OUT} = 50mA	- - -	92 75 55	- - -	dB dB dB
Output Noise Voltage	e_N	10Hz to 100kHz, I_{OUT} = 10mA	-	9.5	-	μV_{RMS}
EN Logic High Threshold	V_{IH}	V_{IN} = 1.1V to 5.5V	1.0	-	-	V
EN Logic Low Threshold	V_{IL}	V_{IN} = 1.1V to 5.5V	-	-	0.4	V
Auto-Discharge Resistance	R_{DIS}	V_{EN} = 0V	-	60	120	Ω
Thermal Shutdown	T_{SD}	Junction temperature rising	-	160	-	°C
Thermal Shutdown Hysteresis	ΔT_{SD}	-	-	20	-	°C

APPLICATION INFORMATION & CIRCUIT DESIGN IN LIFE_GPS

1. Secondary Low-Dropout Architecture:
In the Life_GPS system, SGM2045-3.3 does not take unregulated battery voltage directly. Instead, its input (Pin 4 IN) is fed directly from the primary nano-quiescent buck regulator (SGM6040D) at a stabilized 3.4V rail. The operational headroom is therefore fixed at ΔV = 3.4V - 3.3V = **0.1V (100mV)**. At the full GNSS subsystem load of 48.5mA (ATGM336H 44mA + ATR2652 LNA 4.5mA), the LDO operates with an ultra-low dropout of ~16mV, and the linear power dissipation is strictly limited to 0.1V × 48.5mA = **4.85mW**, eliminating battery-level thermal issues.

2. Switch-Mode Ripple Isolation:
With 92dB PSRR at 1kHz and >55dB beyond 100kHz, the SGM2045 completely absorbs the 1.5MHz switching artifacts from the upstream SGM6040D buck, supplying pristine 9.5μV_{RMS} analogue power to the ATGM336H GNSS receiver and the ATR2652 LNA.

3. Safe Power Sequencing:
Controlled via MCU GPIO P0.04 (GNSS_LDO_EN). Before pulling EN low, the MCU tri-states (High-Z) UART TXD and PPS lines to prevent parasitic leakage through internal ESD protection diodes. When disabled, the internal 60Ω pull-down actively discharges the GNSS VCC line to zero.

ORDERING INFORMATION

Order Model	Output Voltage	Package Description	Package Marking	Packing Option
SGM2045-3.3XXDH4G/TR	3.3V Fixed	XTDFN-1×1-4L (Green)	G3XX	Tape and Reel, 10000
SGM2045-1.8XXDH4G/TR	1.8V Fixed	XTDFN-1×1-4L (Green)	G4XX	Tape and Reel, 10000

SGM2045-3.0XXDH4G/TR	3.0V Fixed	XTDFN-1×1-4L (Green)	G5XX	Tape and Reel, 10000